

Temporally Constrained Segmentation Enables Robust Quantification of Scratch-Wound Healing Dynamics

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Abstract

Objective: To develop and validate a fully automated, training-free computational pipeline for quantitative analysis of scratch-wound healing assays that overcomes limitations of existing methods, including single-frame independence, reliance on annotated training data, and lack of biologically informed temporal constraints.

Materials and Methods: We designed a computational pipeline for phase-contrast time-lapse microscopy that segments wound boundaries using local variance-based texture analysis and enforces a biologically motivated monotonic closure constraint over time. The method operates in $O(HW)$ per frame and requires no training data. The single-frame detector consists of seven stages: local variance mapping, Y-band detection, column-wise edge extraction, RANSAC-based outlier rejection, Savitzky-Golay smoothing, vectorized mask reconstruction, and geometric validation. A stateful temporal integrator constrains per-column boundary evolution to inward motion only. The pipeline was evaluated on 325 images from two independent cell lines (iMC ISOR544C and iMC MUTR544C) across pharmacologic conditions including ALK5 inhibitor, DMSO, and media control at concentrations of 0.1mM.

Results: Expert review confirmed 98.9% detection accuracy across 285 wound-bearing images, with a mean Dice coefficient of 0.91 against manual annotations ($n=5$). The automated quality-control module achieved sensitivity of 1.00 for wound detection. Analysis of 282 validated trajectories demonstrated modest wound closure across all conditions (15.5–20.0% at 24 hours), with ALK5 inhibition yielding the lowest fractional closure ($15.5 \pm 0.8\%$), consistent with suppression of migration. No condition reached 50% closure within the observation window. Computational efficiency enabled processing of a 20-frame trajectory in under 10 seconds on standard hardware.

Discussion: This approach eliminates dependence on training data while incorporating biologically grounded temporal constraints, addressing key limitations of deep learning and frame-independent methods. The resulting trajectories are smooth, physically consistent, and directly interpretable without post hoc correction, enabling robust downstream quantitative analysis.

Conclusion: The proposed pipeline provides a scalable, accurate, and interpretable framework for automated wound healing quantification, bridging a critical methodological gap and supporting reproducible analysis in cell migration studies.

1 Introduction

1.1 The Scratch-Wound Assay

The scratch-wound (or wound-healing) assay is one of the most widely used *in vitro* methods for studying collective cell migration [1–3]. A confluent cell monolayer is mechanically disrupted — typically by drawing a pipette tip across the surface — creating a cell-free gap. Time-lapse microscopy then records the progressive closure of this gap as cells at the wound margin proliferate and migrate inward. The rate and completeness of wound closure serve as proxies for migratory capacity and are routinely used to evaluate drug effects, gene knockdowns, and phenotypic differences between cell lines [4–6]. The assay’s simplicity, low cost, and compatibility with high-content screening platforms have made it a workhorse of cell biology for over two decades [7].

1.2 Quantification Remains a Bottleneck

Despite the assay’s conceptual simplicity, quantification remains a significant bottleneck. Manual measurement of wound area from exported images introduces inter-observer variability, is labor-intensive — especially for multi-well, multi-timepoint experiments — and does not scale to the hundreds of trajectories generated by modern high-content screening platforms. Several computational tools have been proposed, including TScratch [8], ImageJ plugins [9], the Robust Quantitative Scratch Assay [10], AIM [11], CellProfiler pipelines [12], and more recently level-set methods with spatial sector analysis [13]. However, they share common limitations.

Single-frame independence. Most tools process each image in isolation, ignoring the temporal structure of the experiment and producing noisy area-versus-time curves with non-biological fluctuations — apparent wound re-opening that contradicts the physical reality of the assay [14]. This is particularly problematic when computing derived metrics such as closure speed, where non-monotonic noise propagates into erroneous rate estimates.

Training data requirements. Machine-learning approaches, including U-Net-based architectures [15–17] and foundation models such as the Segment Anything Model [18], require annotated ground-truth masks. These annotations are themselves subject to observer bias and are expensive to produce at scale. Deep-learning pipelines further introduce domain-shift risks when applied to imaging conditions different from their training distribution [19].

Lack of biological constraints. No existing tool enforces the physical reality that a properly conducted scratch wound can only close over time. While post-hoc smoothing can reduce noise, it does not encode the underlying biological axiom and may introduce artifacts of its own.

1.3 Biological Context: Myoblast Migration Under Pharmacological Modulation

Cell migration and proliferation are fundamental processes governing tissue repair, development, and pathological states including fibrosis and metastatic dissemination [20, 21]. In skeletal muscle, satellite cell activation, migration, and differentiation are critical for regeneration following injury

[22–24]. The scratch-wound assay provides a reproducible *in vitro* model for studying these processes in myoblasts [25].

Two pharmacological targets are of particular interest in the context of skeletal muscle repair.

The renin–angiotensin system (RAS). Angiotensin II (AngII), acting through the AT1 receptor (AT1R), exerts pleiotropic effects on skeletal muscle biology, including inhibition of satellite cell proliferation [26], regulation of satellite cell chemotaxis [27], and promotion of myofibroblast transdifferentiation [28]. AT1R blockade with losartan was shown to attenuate TGF- β -induced failure of muscle regeneration in multiple myopathic states [29] and to improve muscle regeneration in non-dystrophic injury models [30]. These findings position AT1R blockers (ARBs) as potential modulators of myoblast migration, though the effects are context-dependent: AT1R deficiency in skin wound models delays healing and reduces angiogenesis [31]. Candesartan, a selective AT1R antagonist, was selected for this study as a clinically available ARB with well-characterized pharmacology [32].

The TGF- β /ALK5 pathway. Transforming growth factor- β (TGF- β) signaling through ALK5 (TGF- β receptor type I) plays a dual role in skeletal muscle: it drives myogenic-to-fibrotic cell fate switching [33, 34], suppresses myoblast differentiation via Smad3-mediated repression of myogenic transcription factors [35], and curbs cell fusion during regeneration [36]. Pharmacological ALK5 inhibition has been shown to attenuate muscle fibrosis in dystrophic models [29, 37] and cancer cachexia [38], and recent work has systematically compared TGF- β receptor I inhibitors for their effects on myotube formation [39] and myofibroblast differentiation [40]. Notably, TGF- β also influences collective cell migration by modulating leader cell positioning at the wound edge [41], suggesting that ALK5 inhibition could directly impact wound closure dynamics.

Critically, AngII and TGF- β signaling pathways converge at the Smad2/3 level: AngII activates Smad3 independently of TGF- β ligand [42, 43], and both pathways synergize to promote fibrosis [44]. This crosstalk provides the rationale for testing both AT1R blockade and ALK5 inhibition in the same experimental framework.

1.4 Our Contribution

We introduce **Quantification**, a two-stage pipeline that addresses the limitations of existing wound quantification tools:

1. A **variance-based single-frame detector** that exploits the fundamental texture difference between cell monolayer (high local variance due to cell edges, organelles, and phase-contrast halos) and the wound channel (smooth, low variance). This approach is grounded in the physics of phase-contrast image formation [45] and follows the precedent of training-free texture-based segmentation in label-free microscopy [46, 47]. No training data or threshold tuning by the user is required.
2. A **stateful temporal integrator** that encodes the monotonic closure axiom directly into the

segmentation: once a pixel column has been declared healed, it cannot revert. This eliminates non-biological noise in the area-versus-time curve and produces smooth, strictly non-increasing wound trajectories. While temporal constraints have been explored in cell tracking [48–50] and video segmentation [51], to our knowledge this is the first application of a hard monotonic constraint to wound-healing segmentation.

The pipeline is complemented by automated multi-level quality control, parallel trajectory processing, and downstream metric extraction (closure speed, edge displacement, closing fraction). We validate Quantification on a pharmacological study of wound closure in immortalized myoblasts treated with candesartan and an ALK5 inhibitor, demonstrating 98.9% detection accuracy against expert annotation and biologically interpretable results across 282 trajectories.

2 Materials and Methods

2.1 Cell Culture and Experimental Design

A full factorial design was implemented, comprising 2 cell lines $\times$ 3 treatment conditions $\times$ 2 independent experiments, yielding 12 unique experimental conditions. Two independent experiments were conducted as biological replicates: EXP1 ($n = 158$ wells) and EXP2 ($n = 167$ wells), resulting in a total of 325 wound regions.

The ALK5 inhibitor was applied at 0.1 mM as a selective antagonist of the TGF- β receptor type I (ALK5). Inhibition of TGF- β signaling is relevant in this context as it promotes myogenic-to-fibrotic transdifferentiation, suppresses myoblast differentiation, and regulates leader cell dynamics during collective migration. DMSO was used as the vehicle control, while untreated media served as the basal reference condition.

Confluent monolayers were wounded using a standardized mechanical scratch assay. Treatment media were applied immediately after wounding. Time-lapse imaging began at $t = 0$ (baseline, fully open wound) and continued for approximately 24 hours, with images acquired at ~ 1 -hour intervals, yielding 20–25 timepoints per trajectory. Images were captured at a resolution of approximately 900×735 pixels per field of view.

iMCs were seeded into 96-well plates at a density of 5×10^4 cells per well in media containing either DMSO or the indicated drug conditions. After overnight incubation, wounds were created in the cell monolayers using the Incucyte[®] 96-Well WoundMaker tool. Cells were washed twice with PBS, after which conditioned media was reapplied. Images were acquired immediately after wounding and every hour for up to 24 hours using the Incucyte[®] Live-Cell Analysis System with the whole-well (4 $\times$) setting.

2.2 Automated Wound Segmentation

The quantification pipeline operates in two stages: (1) a single-frame detector that extracts wound boundaries from each image independently, and (2) a temporal integrator that enforces the monotonic closure constraint across the time series. Parameters were selected based on empirical tuning on a held-out subset and fixed for all analyses.

For each trajectory, frames are processed sequentially as follows: each frame is independently segmented using the single-frame detector (Section 2.2.2), producing validated edge estimates; these are refined by the temporal integrator (Section 2.2.3) under a monotonic constraint; the resulting edges are converted to wound masks and areas.

2.2.1 Problem Formulation

Let $I(x, y, t) \in [0, 1]$ denote the intensity-normalized grayscale image at spatial position (x, y) and discrete timepoint $t \in \{0, 1, \dots, T\}$. The image has dimensions $H \times W$ (rows $\times$ columns). We seek a binary wound mask:

$$M(x, y, t) = \begin{cases} 1 & \text{if pixel } (x, y) \text{ is wound at time } t \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

and per-column edge functions:

$$u(x, t) = \min\{y : M(x, y, t) = 1\} \quad (\text{upper edge}) \quad (2)$$

$$\ell(x, t) = \max\{y : M(x, y, t) = 1\} \quad (\text{lower edge}) \quad (3)$$

The wound area at time t is:

$$A(t) = \sum_{x=0}^{W-1} \max(0, \ell(x, t) - u(x, t) + 1) \quad (4)$$

The biological axiom (monotonic closure) for wound area is a monotonically non-increasing function of time:

$$\boxed{A(t+1) \leq A(t) \quad \forall t \geq 0} \quad (5)$$

2.2.2 Single-Frame Detection

The single-frame detector maps an input image $I(x, y, t)$ to validated edge functions $u_{\text{det}}(x, t)$ and $\ell_{\text{det}}(x, t)$ through a seven-stage pipeline that transforms a raw grayscale image into a geometrically consistent wound mask with smooth boundary curves (Figure 1).

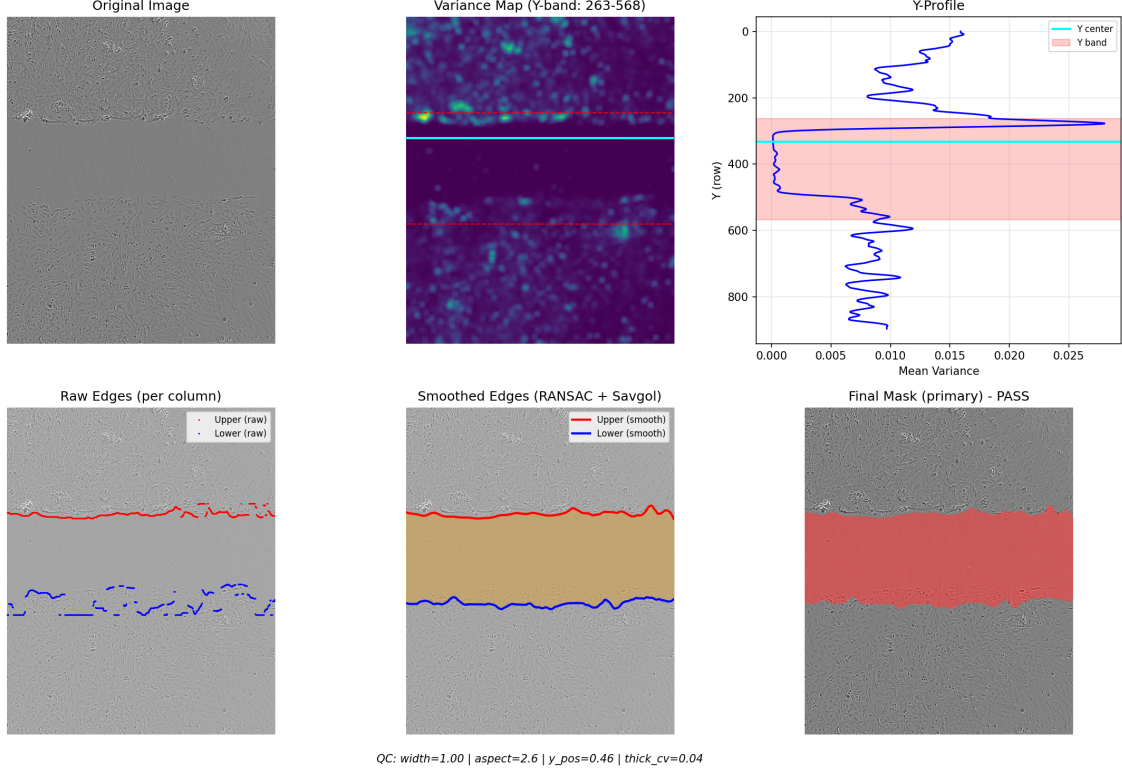


Figure 1: Representative pipeline output on a single frame. Phase-contrast image acquired at $t = 0$ (ALK5i condition, EXP2). The wound channel (center, low local variance) is flanked by cell monolayers (high local variance). Detected upper and lower boundaries are overlaid as colored lines after RANSAC outlier rejection and Savitzky–Golay smoothing. The shaded region corresponds to the reconstructed binary wound mask (Equation (7)).

Stage 1: Local Variance Texture Map. Cell monolayers exhibit high spatial variance due to cell borders, nuclei, and phase-contrast halos, while the wound channel is comparatively smooth [45] (Figure 1b). We use local variance as a texture discriminator [46, 52, 53]. For a square window of side length w (default $w = 15$), the local variance at each pixel is computed via the uniform-filter identity:

$$\sigma^2(x, y) = \langle I^2 \rangle_w(x, y) - [\langle I \rangle_w(x, y)]^2 \quad (6)$$

where $\langle \cdot \rangle_w$ denotes the box-filter mean over a $w \times w$ neighborhood, yielding $O(HW)$ complexity independent of window size.

Stage 2: Y-Band Detection. The wound is identified as a horizontal band spanning nearly the full image width. We compute the row-averaged variance profile $\bar{V}(y)$, smooth it with a 1-D Gaussian kernel ($\sigma_y = 10.0$ pixels), locate the global minimum y_c , and extract the contiguous segment of sub-threshold rows (35th percentile) containing y_c , expanded by a margin fraction of 0.15.

Stage 3: Column-Wise Edge Extraction. Within the Y-band, upper and lower wound boundaries are extracted for each column. The edge-detection threshold τ is derived adaptively from the cell regions flanking the wound: $\tau = 0.5 \cdot \text{median}(\sigma_{\text{cell}}^2)$. Edge search is fully vectorized using

array boolean operations.

Stage 4: RANSAC Outlier Rejection. We apply an iterative polynomial-fit-and-reject scheme inspired by RANSAC [54]. A degree-2 polynomial captures genuine wound curvature while rejecting localized spikes. Three iterations with a maximum deviation threshold of 25 pixels are used.

Stage 5: Savitzky–Golay Smoothing. Residual high-frequency jitter is removed with a Savitzky–Golay filter [55], preserving edge shape without phase distortion. Parameters: window length $w_{\text{sg}} = 51$, polynomial order $p_{\text{sg}} = 3$.

Stage 6: Mask Reconstruction. The binary mask is reconstructed from smooth edges using vectorized broadcasting:

$$M(x, y) = [u(x) \leq y \leq \ell(x)] \quad (7)$$

Stage 7: Geometric Validation. Four quality-control criteria are evaluated on each mask (Table 1). A detection passes QC if and only if all four criteria are satisfied. When the primary pipeline fails, an optional fallback strategy is activated using constrained region growing from a seed at y_c with morphological cleanup (hole filling at 500 px, small-object removal at 1000 px, disk closing at radius 5).

Table 1: Geometric validation criteria applied to each wound mask.

Criterion	Threshold	Rationale
Width coverage r_w	> 0.75	Wound must span most of image width
Aspect ratio ρ	> 1.5	Wound is wider than tall
Y-position $\bar{y}_{\text{frac}}$	$\in (0.15, 0.85)$	Wound is roughly centered
Thickness CV	< 0.60	Consistent thickness across columns

2.2.3 Temporal Constraint: Kalman-Filtered Monotonic Closure

A naive implementation of the monotonic closure constraint enforces a hard per-column clamp, such that the upper edge $u(x, t)$ can only move inward relative to $u(x, t - 1)$ (and symmetrically for the lower edge). While this deterministic rule strictly encodes the biological axiom (Equation 5), it introduces an irreversible failure mode: spurious inward detections—arising from debris, illumination artifacts, or transient focus shifts—are permanently propagated to all subsequent frames, with no mechanism for recovery.

To mitigate this limitation, we replace the hard clamp as the primary update mechanism with a per-column Kalman filter. The filter fuses the detector output with a monotonic prior, weighting each according to a data-driven estimate of observation uncertainty. Noisy or low-contrast detection are down-weighted, while consistent high-confidence observations are retained.

A hard monotonic constraint is subsequently applied as a secondary safety mechanism to guarantee $A(t) \leq A(t - 1)$ under all conditions. With this hierarchical design the Kalman filter handles typical

artifact-contaminated observations, while the hard constraint provides an unconditional correctness for residual failure cases.

The full per-frame processing sequence of the temporal integrator, including detector input, filtering steps, constraint enforcement, and numerical safeguards, is summarized in Supplementary Table S3. Four numerical safeguards (Supplementary Table S3, Steps 2, 7, 9, 11) prevent failure-mode propagation across frames; their formal statement follows below.

At each timepoint t , the single-frame detector (Section 2.2.2) produces per-column edge estimates $u_{\text{det}}(x, t)$ and $\ell_{\text{det}}(x, t)$, corresponding to the upper and lower wound boundaries, respectively. These estimates are treated as observations and are mapped by the temporal integrator to constrained edges $u(x, t)$ and $\ell(x, t)$ that satisfy the monotonic closure constraint. We denote by $P(x, t)$ the per-column uncertainty associated with the temporal estimate.

At the initial timepoint $t = 0$, the detector outputs are accepted without constraint:

$$u(x, 0) = u_{\text{det}}(x, 0), \quad \ell(x, 0) = \ell_{\text{det}}(x, 0) \quad (8)$$

These edges define the reference wound geometry for the trajectory. The corresponding uncertainty is initialized as

$$P(x, 0) = \max(R_{\text{base}}(0), 1.0) \quad (9)$$

where $R_{\text{base}}(0)$ is a baseline estimate of the single-frame detector noise variance at $t = 0$ (px^2), computed from the spatial residuals between the RANSAC-rejected edge (Stage 4, Section 2.2.2) and the Savitzky–Golay smoothed edge (Stage 5):

$$R_{\text{base}}(0) = \text{Var}_x(u_{\text{RANSAC}}(x, 0) - u_{\text{det}}(x, 0)) \quad (10)$$

where $u_{\text{RANSAC}}(x, 0)$ denotes the per-column edge after RANSAC outlier rejection but before smoothing, and $u_{\text{det}}(x, 0)$ is the final Savitzky–Golay smoothed output. This quantity captures the within-frame spatial variability of the edge estimate at baseline: large residuals indicate an irregular or noisy wound boundary, yielding higher initial uncertainty and a weaker filter dependence on the first-frame measurement.

If edge residuals are unavailable (fewer than six finite-valued columns, or fallback detection at $t = 0$), $R_{\text{base}}(0)$ falls back to $0.1 \times Q = 40 \text{ px}^2$ (using the default $Q_0 = 400 \text{ px}^2$). The floor $\max(\cdot, 1.0)$ in Equation (9) prevents zero initial uncertainty, which would collapse the Kalman gain to zero on the first update step and make the filter unresponsive to all subsequent observations.

Kalman-filtered constraint ($t > 0$). For subsequent timepoints, the temporal integrator updates the constrained edges by combining the detector observations with the previous state under a monotonicity prior. The update proceeds in three steps.

Prediction. The predicted state is the previous constrained edge; predicted uncertainty grows by

process noise Q :

$$\hat{u}^-(x, t) = u(x, t - 1), \quad P^-(x, t) = P(x, t - 1) + Q \quad (11)$$

Process noise Q (px²) is estimated empirically from QC-validated trajectories $\mathcal{C}$ as the variance of consecutive frame-to-frame edge differences:

$$\hat{Q} = \text{Var}\left(\{u_{\text{det}}(x, t+1) - u_{\text{det}}(x, t) \mid x, t \in \mathcal{C}\}\right) \quad (12)$$

A conservative default $Q_0 = 400$ px² is used on first run.

Spatial pre-smoothing. The raw detected edge is smoothed with a 1-D Gaussian kernel ($\sigma_{\text{spatial}} = 5$ columns) before the Kalman update to suppress column-level artefacts:

$$\tilde{u}_{\text{det}}(x, t) = (G_{\sigma} * u_{\text{det}}(\cdot, t))(x) \quad (13)$$

Per-column observation noise. Observation noise $R(x, t)$ is estimated from three data-driven signals:

$$R(x, t) = R_{\text{base}}(t) \cdot C^{-2}(x, t) \cdot J(x, t) \cdot m_{\text{method}} \quad (14)$$

where $R_{\text{base}}(t) = \text{Var}_x(u_{\text{det}} - \tilde{u}_{\text{det}})$ is the frame-level detector noise, $C(x, t)$ is the local wound-vs-cell variance contrast (low contrast implies high R), $J(x, t)$ is a jump penalty for large deviations from the prediction, and $m_{\text{method}} \in \{1.0, 3.0\}$ penalises the less reliable fallback detector.

Kalman update. The gain $K(x, t) \in [0, 1]$ blends prediction and observation:

$$K(x, t) = \frac{P^-(x, t)}{P^-(x, t) + R(x, t)} \quad (15)$$

$$u_{\text{filt}}(x, t) = u(x, t - 1) + K(x, t)[\tilde{u}_{\text{det}}(x, t) - u(x, t - 1)] \quad (16)$$

$$P_{\text{upd}}(x, t) = (1 - K(x, t)) \cdot P^-(x, t) \quad (17)$$

High R (noisy observation) implies small K , causing the filter to rely on the monotonic prior. Low R (clean, high-contrast frame) implies large K , trusting the detector.

Hard safety net (secondary guarantee). After the Kalman blend, the biological invariant is enforced unconditionally:

$$u(x, t) = \max(u_{\text{filt}}(x, t), u(x, t - 1)) \quad (18)$$

$$\ell(x, t) = \min(\ell_{\text{filt}}(x, t), \ell(x, t - 1)) \quad (19)$$

This guarantees $A(t) \leq A(t - 1)$ for all $t > 0$ regardless of Kalman output.

The pipeline ultimately produces, for each trajectory, a sequence of wound masks and the corresponding area time series $A(t)$. The Kalman process adds $O(W)$ operations per frame. Total per-frame complexity remains $O(HW)$, dominated by the variance map computation.

Wound Closure. When the normalized wound area satisfies $\frac{A(t-1)}{A(0)} < \theta_{\text{close}}$ with $\theta_{\text{close}} = 0.01$ the wound is considered fully closed. For all subsequent timepoints, the edge positions are held constant $u(x, t) = u(x, t - 1)$ and $\ell(x, t) = \ell(x, t - 1)$, preventing numerical fluctuations after closure.

Four safeguards prevent numerical failure from propagating across frames: (i) non-finite observation noise R is clamped to $Q_0 = 400 \text{ px}^2$; (ii) non-finite Kalman filter outputs are reset to the previous constrained edge; (iii) per-column uncertainty P is reset to P^- when the safety net overrides the Kalman estimate, preventing overconfidence propagation; and (iv) a last-resort area comparison emits a warning and reverts to the previous frame if $A(t) > A(t - 1)$ persists after all constraint paths.

2.3 Wound Quantification

Once the wound boundaries have been extracted and temporally constrained, quantitative measures of wound healing can be derived from the resulting area trajectory $A(t)$ and edge positions $u(x, t)$ and $\ell(x, t)$. These metrics represent the inward movement of the wound boundaries at each column x .

Fractional wound closure is defined as the proportion of the initial wound area that has closed over time. It ranges from 0 (no closure) to 1 (complete closure), and is given by

$$f(t) = \frac{A(0) - A(t)}{A(0)}. \quad (20)$$

To quantify spatial dynamics, we measure the displacement of the upper and lower wound edges between the initial timepoint $t = 0$ and a target time t_T defining the edge displacement as :

$$\Delta u(x) = u(x, t_T) - u(x, 0), \quad (21)$$

$$\Delta \ell(x) = \ell(x, 0) - \ell(x, t_T). \quad (22)$$

Wound healing rate is defined by converting edge displacements into velocities through normalization by elapsed time. The mean closure speed represents the average inward movement of the wound boundaries and is defined as

$$v_{\text{mean}} = \frac{v_{\text{upper}} + v_{\text{lower}}}{2}, \quad (23)$$

where each edge velocity is computed as the average displacement per unit time across valid columns:

$$v_{\text{edge}} = \frac{1}{|\mathcal{V}|} \sum_{x \in \mathcal{V}} \frac{|\Delta_{\text{edge}}(x)|}{t_T}. \quad (24)$$

This metric provides a summary of the overall migration rate while reducing the influence of local irregularities along the wound boundary.

2.4 Statistical Analysis

Pairwise treatment comparisons versus DMSO were performed using Welch’s t -test (no equal-variance assumption), stratified by cell line. Multiple comparison correction was applied using the Bonferroni method. Effect sizes were quantified as Cohen’s d defined as $d = \frac{\bar{x}_{\text{treated}} - \bar{x}_{\text{control}}}{s_{\text{pooled}}}$.

The time-series analysis was performed using a linear mixed-effects model. The response variable distance_{ij} for trajectory i at timepoint j was modeled as

$$\text{distance}_{ij} = \mu + \beta_1 t_j + \beta_2 \text{condition}_i + \beta_3 \text{cell_line}_i + \beta_4 (t_j \times \text{condition}_i) + \beta_5 (t_j \times \text{cell_line}_i) + b_{0i} + b_{1i} t_j + \varepsilon_{ij}.$$

Here, μ denotes the fixed intercept, β_k the fixed-effect coefficients, b_{0i} and b_{1i} the random intercept and slope for trajectory i , and ε_{ij} the residual error term.

Model assumptions were evaluated by testing residual normality using the Shapiro–Wilk test and homogeneity of variance using Levene’s test (see Supplementary Table S8).

2.5 Implementation

The pipeline was implemented in Python 3.11 using NumPy, OpenCV, SciPy and scikit-image [56]. The pipeline is fully deterministic and reproducible. The per-frame computational complexity is $O(HW)$, dominated by the variance map computation. For a typical 1024×1280 image, this corresponds to approximately 1.3×10^6 operations per stage.

3 Results

3.1 Image Census and Exclusions

A total of 325 images at $t = 0$ images were processed across two exposure conditions (Candesartan, $n = 191$; ALK5i, $n = 134$) and two independent experiments (EXP1, $n = 158$; EXP2, $n = 167$).

Manual review identified **40 images (12.3%)** containing no visible wound, which were excluded from accuracy analysis. The no-wound images were unevenly distributed (Table 2): 38 of 40 occurred in EXP1. Two images remained unannotated. The remaining **285 wound-bearing images** formed the validation cohort.

Table 2: Distribution of no-wound images by condition and experiment. These images contained no visible scratch wound and were excluded from accuracy analysis.

Condition	EXP1	EXP2	Total
Candesartan	29	1	30
ALK5i	9	1	10
Total	38	2	40

3.2 Wound Detection Accuracy

Of 285 wound-bearing images, the algorithm correctly segmented **282 (98.9%)** as judged by expert review. Only 3 images were incorrectly segmented. Performance was consistent across conditions and experiments (Table 3).

Table 3: Detection accuracy by exposure and experiment (wound-bearing images only). Values are expressed as correctly segmented / total (percentage). Expert review served as ground truth.

Condition	EXP1	EXP2	Total
Candesartan	67/67 (100.0%)	93/94 (98.9%)	160/161 (99.4%)
ALK5i	51/53 (96.2%)	71/71 (100.0%)	122/124 (98.4%)
Total	118/120 (98.3%)	164/165 (99.4%)	282/285 (98.9%)

3.3 Automatic Quality-Control Evaluation

The automatic QC flag was evaluated against human verdicts in two contexts.

All annotated images ($n = 323$). The QC algorithm achieved 94.4% agreement with the human reviewer (Cohen’s $\kappa = 0.69$; Table 4). It identified all 282 correctly segmented images (sensitivity = 1.00, NPV = 1.00) but flagged only 23 of 41 problematic images as invalid (specificity = 0.56). The 18 false positives (QC valid, human incorrect) were predominantly no-wound images (16 of 18) that the variance-based QC could not distinguish from valid segmentations.

Wound-bearing images only ($n = 285$). The QC algorithm correctly classified 283 of 285 images (accuracy = 99.3%), missing 2 incorrect segmentations while producing no false negatives.

Table 4: QC confusion matrix (all annotated images, $n = 323$). Sensitivity = 1.00; Specificity = 0.56; PPV = 0.94; NPV = 1.00.

	Human: Correct	Human: Incorrect
QC: Valid	282 (TP)	18 (FP)
QC: Invalid	0 (FN)	23 (TN)

3.4 Pixel-Level Segmentation Quality

For 5 images with expert-drawn polygon annotations, pixel-level metrics were computed against human ground truth (Table 5). The mean Dice coefficient was 0.91 (SD = 0.06), indicating high spatial overlap. The high recall (0.97) relative to precision (0.87) indicates a tendency to slightly over-segment (include background pixels) rather than miss wound area.

Table 5: Pixel-level segmentation metrics against expert polygon annotations ($n = 5$). Dice and F1 are identical for binary masks. Hausdorff distance is reported at the 95th percentile.

Metric	Mean	SD	Min	Max
Dice	0.909	0.057	0.843	0.998
IoU	0.837	0.099	0.729	0.997
Precision	0.868	0.123	0.729	1.000
Recall	0.969	0.070	0.844	1.000
F1	0.909	0.057	0.843	0.998
Accuracy	0.954	0.029	0.921	0.999
Hausdorff 95 (px)	45.6	27.8	0.0	76.0

3.5 Wound Healing Dynamics

3.5.1 Validated Cohort and Pipeline Illustration

Of the 282 verified segmentations, all were included in wound healing dynamics analysis: DMSO ($n = 100$), Candesartan ($n = 86$), ALK5i ($n = 55$), Media ($n = 41$). Two cell lines were represented: iMC ISOR544C ($n = 145$) and iMC MUTR544C ($n = 137$). Each trajectory comprised 20–25 timepoints over approximately 24 hours. A representative pipeline output is shown in Figure 2.

3.5.2 Fractional Closure at Fixed Timepoints

Mean fractional closure at 24 hours ranged from 15.5% (ALK5i) to 20.0% (Candesartan), with DMSO at 19.7% and Media at 18.5% (Table 6). At 12 hours, the pattern was similar: ALK5i showed the lowest mean closure (12.0%) and DMSO/Candesartan were comparable (14.5% and 14.9%; Table 7).

Table 6: Mean fractional wound closure at 24 hours by treatment condition (mean $\pm$ SEM, all concentrations pooled). No condition reached 50% closure within the observation window.

Condition	n	Mean Closure	SEM
DMSO	100	0.197	0.010
Candesartan	86	0.200	0.011
ALK5i	55	0.155	0.008
Media	41	0.185	0.012

Table 7: Mean fractional wound closure at 12 hours by treatment condition (mean $\pm$ SEM, all concentrations pooled).

Condition	n	Mean Closure	SEM
DMSO	100	0.145	0.006
Candesartan	86	0.149	0.009
ALK5i	55	0.120	0.007
Media	41	0.131	0.008

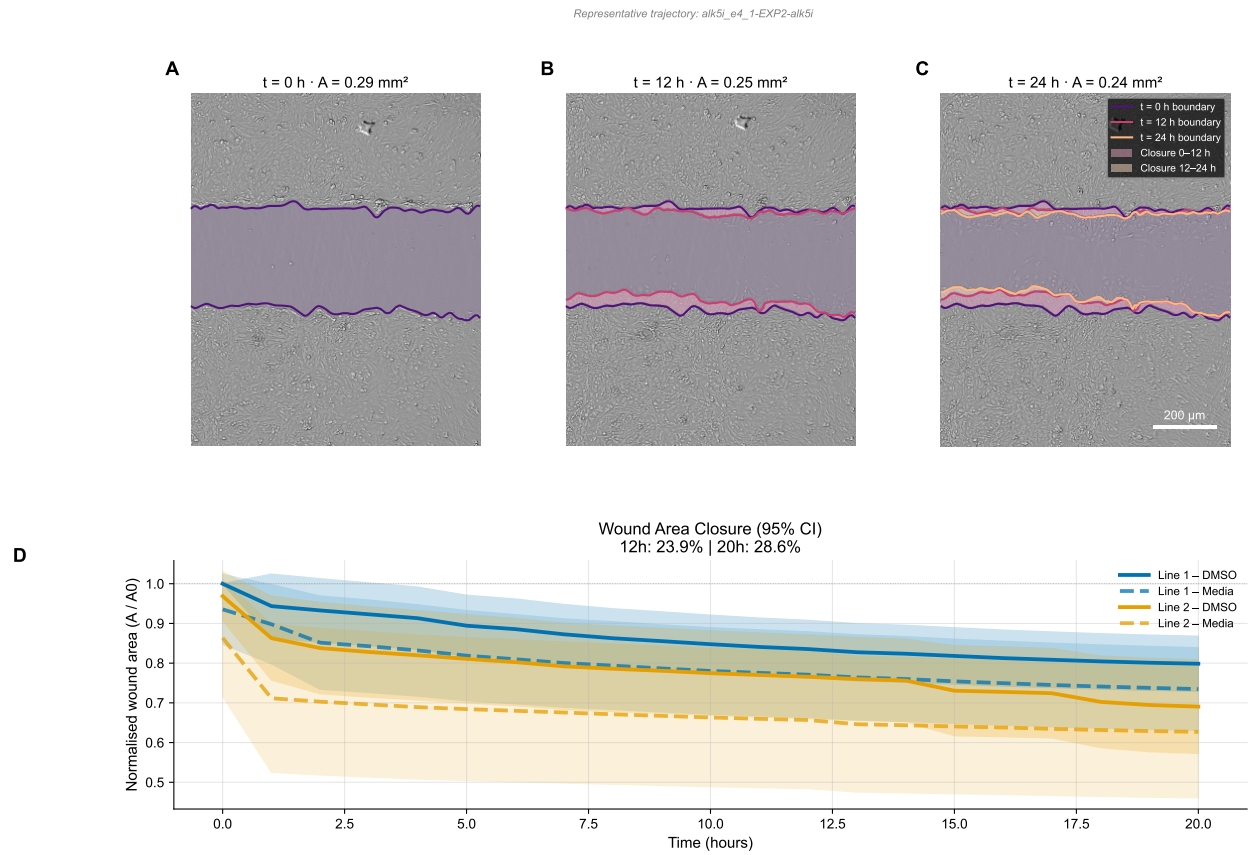


Figure 2: Automated wound boundary detection and model quality metrics. (A–C) Representative phase-contrast images of a wound healing assay at three timepoints ($t = 0$, mid, and $t = 20$ h) with computationally detected wound boundaries overlaid as semi-transparent filled regions. Earlier timepoints are shown as faint dashed outlines; the current timepoint as solid fill. Images were acquired at $1.24 \mu\text{m}/\text{px}$. (D) Normalised wound area over time, expressed as a fraction of the $t = 0$ wound area. Error bands represent 95% confidence intervals. Sample sizes per group are provided in the legend. (Figure 2 corresponds to the analysis of the ALK5i exposure, one cell line subset; full dataset is shown in Figure 3.)

3.5.3 Time to 50% Closure

No condition reached 50% closure within the 24-hour observation window.

3.5.4 Time-Course Profiles and Cross-Sectional Analysis

Wound closure trajectories followed a characteristic pattern across all conditions (Figure 3): an initial lag phase (0–4 h) with minimal closure, followed by a progressive closure phase that did not plateau within 24 hours. The ALK5i-treated group showed the slowest kinetics throughout the observation period, separating from other conditions by approximately 6–8 hours post-wounding. DMSO and candesartan groups showed similar kinetic profiles, while the media control fell between ALK5i and the vehicle-treated groups.

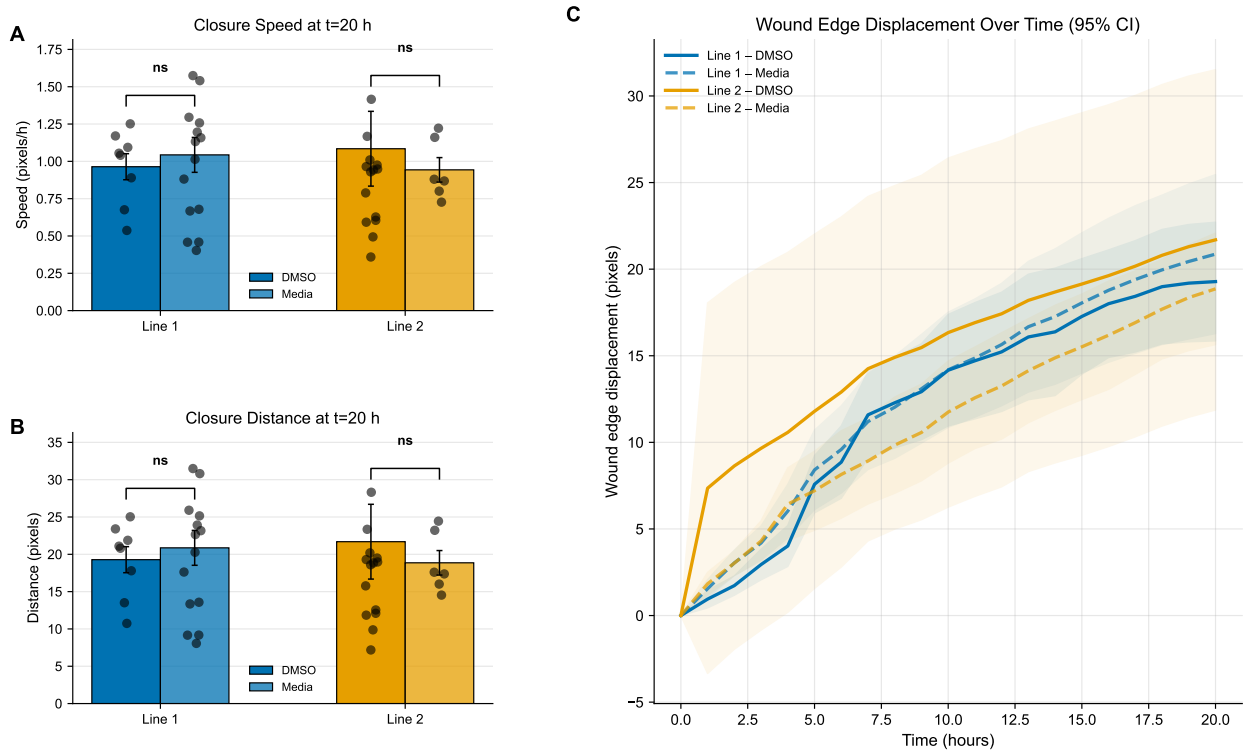


Figure 3: Wound healing dynamics across experimental conditions. (A) Mean wound closure speed (pixels/h) at $t = 20$ h, computed as the rate of wound edge displacement. (B) Mean wound closure distance (pixels) at $t = 20$ h, representing cumulative inward displacement of the wound boundary from $t = 0$. (C) Time-series of wound edge displacement over the full observation period. Error bars represent 95% confidence intervals. Statistical comparisons were performed using Welch's t -test with Bonferroni correction. Sample sizes: Line 1 – Alk5i: $n = 10$; Line 1 – DMSO: $n = 8$; Line 1 – Media: $n = 15$; Line 2 – Alk5i: $n = 5$; Line 2 – DMSO: $n = 15$; Line 2 – Media: $n = 6$.

Sample-level accounting, QC pass rates, and full descriptive statistics are provided in Supplementary Tables S1, S2, and S5. Inter-experiment reproducibility data are reported in Supplementary Table S9. Full pairwise comparison results including confidence intervals appear in Supplementary Table S6. The full descriptive statistics per group are in Supplementary Table S4.

Cross-sectional analysis at the final timepoint confirmed no statistically significant differences between ALK5i and DMSO after Bonferroni correction across both cell lines (Line 1: $p = 0.38$, Cohen’s $d = -0.40$; Line 2: $p = 0.40$, Cohen’s $d = -0.29$; Table 8). Media did not differ significantly from DMSO in either cell line ($p > 0.59$ in both cases). These results are consistent with modest absolute differences that are underpowered given current sample sizes.

Table 8: Effect sizes and pairwise comparisons vs. DMSO at the final timepoint. Welch’s t -test with Bonferroni correction; p -values reported after correction. All comparisons non-significant after correction.

Cell line	Treatment	Control mean	Treated mean	Difference	Cohen’s d	p (corrected)	n_{ctrl}	n_{trt}
Line 1	Alk5i	19.28	16.34	−2.94	−0.402	1.000	8	10
Line 1	Media	19.28	20.87	1.59	0.200	1.000	8	15
Line 2	Alk5i	21.69	16.56	−5.13	−0.294	1.000	15	5
Line 2	Media	21.69	18.86	−2.83	−0.169	1.000	15	6

Note: Mean distances are in pixels. All p -values = 1.000 after Bonferroni correction (4 comparisons). Cohen’s d is computed using pooled standard deviation.

3.5.5 Mixed-Effects Time-Series Model

The linear mixed-effects model (Equation (??)) confirmed a highly significant main effect of time ($\beta = 0.879$ px/frame, $p < 10^{-12}$), indicating consistent wound closure across all conditions. No time $\times$ condition or time $\times$ cell line interaction reached significance (all $p > 0.31$; Supplementary Table S7), consistent with the cross-sectional analysis. Full mixed-model parameter estimates and variance components are provided in Supplementary Table S7.

4 Discussion

4.1 Methodological Advances

The Quantification pipeline offers several advances over existing scratch-wound quantification tools.

Training-free segmentation. Unlike deep-learning approaches that require annotated datasets [15–18] and suffer from domain shift when applied to novel imaging conditions [19], Quantification exploits a fundamental physical property of phase-contrast image formation: cells scatter light and produce high local variance, while the empty wound channel does not [45]. This approach aligns with growing evidence that classical texture-based methods can compete with machine learning on small-to-medium datasets in label-free microscopy [46, 47]. The 98.9% detection accuracy achieved without any training data is competitive with or exceeds reported performance of supervised methods on comparable assays.

Biologically grounded temporal constraint. The monotonic closure axiom is not merely a regularizer — it encodes a physical law of the assay system. While temporal constraints have been explored in cell tracking [48, 49] and video segmentation [51], prior wound-healing tools process

frames independently. The per-column monotonic constraint is strictly stronger than a global area constraint: it enforces spatial coherence at every column, preventing the non-biological area fluctuations that Treloar and Simpson [14] identified as a major source of quantification error.

Robust edge detection. The combination of RANSAC-inspired outlier rejection [54] with Savitzky–Golay smoothing [55] provides a principled two-stage approach to boundary regularization. The polynomial fit captures genuine wound curvature while rejecting localized artifacts, and the filter preserves edge shape without phase distortion.

Computational efficiency. Full vectorization ensures $O(HW)$ complexity per frame. A typical 20-frame trajectory processes in under 10 seconds on commodity hardware, enabling real-time analysis of high-content screening experiments [4, 57, 58].

Multi-level quality control. The three-tier QC system provides hierarchical confidence in segmentation quality. The perfect sensitivity (1.00) for wound-bearing images ensures that no valid segmentation is erroneously rejected, while the moderate specificity for no-wound images (0.56) represents a known limitation addressable by supplementing variance-based QC with intensity-profile features.

4.2 Biological Findings

The wound healing dynamics observed are consistent with the known biology of the pharmacological targets.

Candesartan showed no significant effect on wound closure. Despite evidence that AngII regulates satellite cell chemotaxis [27] and proliferation [26], AT1R blockade with candesartan did not substantially alter myoblast wound closure kinetics relative to DMSO (20.0% vs. 19.7% at 24 h). This may reflect cell-type-specific differences in AT1R expression or the possibility that pro-migratory and anti-migratory effects of AngII signaling are balanced in this myoblast system. Notably, AT1R deficiency delays wound healing in skin models [31], suggesting context-dependent effects.

ALK5 inhibition reduced wound closure. The ALK5i-treated group showed the lowest fractional closure (15.5% at 24 h), consistent with the role of TGF- β in promoting collective cell migration through leader cell positioning [41] and epithelial-to-mesenchymal transition-related motility pathways. While TGF- β is often considered anti-regenerative in skeletal muscle due to its pro-fibrotic effects [33, 37, 59], the reduced wound closure here suggests that ALK5 signaling also contributes to the migratory component of wound repair in myoblasts. This finding aligns with recent work showing that TGF- β receptor I inhibitors have complex, dose-dependent effects on myoblast biology [39, 40].

No condition reached 50% closure. The observation that all conditions achieved less than 20% closure in 24 hours may reflect the inherently slower migratory phenotype of skeletal muscle myoblasts compared to epithelial cell lines [22, 25], or specific culture conditions. Longer observation windows (48–72 h) may be necessary to fully characterize dose-response relationships, as noted for

other slowly migrating cell types [60].

AngII–TGF- β crosstalk. The differential effects of candesartan (no change) and ALK5 inhibitor (reduced closure) are informative in light of the known convergence of these pathways at Smad2/3 [42–44]. The divergent outcomes suggest that, in this myoblast system, migration-relevant TGF- β signaling operates predominantly through the canonical ALK5–Smad pathway rather than through AngII-mediated Smad activation.

4.3 Comparison with Existing Tools

Table 9 contextualizes Quantification’s properties against established and recent alternatives. Quantification is unique in combining training-free operation with hard temporal constraints and multi-level quality control. While deep-learning methods can achieve high per-frame accuracy when adequate training data are available, Quantification eliminates the domain-shift problem entirely and produces temporally coherent outputs by construction.

Table 9: Comparison of wound-healing quantification tools. DL, deep learning; RQSA, Robust Quantitative Scratch Assay.

Feature	TScratch [8]	ImageJ [9]	RQSA [10]	DL [16]	Level-set [13]	Quantification (this work)
Training data	No	No	No	Yes	No	No
Temporal constraint	No	No	No	No	No	Yes
Automated QC	No	No	Partial	No	No	Yes (3-level)
Spatial analysis	No	Width only	No	Pixel-level	Sector	Per-column
Outlier rejection	No	No	Statistical	Implicit	No	RANSAC
Parallel processing	No	Macro	No	GPU batch	No	CPU parallel

4.4 Limitations

Several known limitations warrant discussion.

Irreversible per-column constraint. If the detector produces a spurious inward edge (due to debris, bubble, or focus artifact), the monotonic constraint propagates that error to subsequent frames. This limitation has been mitigated by the Kalman-filtered constraint (default), which assigns high observation noise to spurious observations. The original hard constraint remains available for reproducibility.

Fixed threshold heuristic. The edge-detection threshold ($\tau = 0.5 \cdot \text{median}(\sigma_{\text{cell}}^2)$) assumes unimodal cell variance. Images with heterogeneous cell populations or illumination gradients may require adaptive thresholding.

Integer-pixel edges. Edges are extracted at integer-pixel resolution. For thin wounds approaching closure, the 1-pixel quantization error represents a significant fraction of remaining width. Sub-pixel edge localization via parabolic interpolation would improve precision.

No uncertainty quantification. The pipeline produces point estimates without confidence intervals. Bootstrapping area estimates under small edge perturbations would provide uncertainty bounds essential for rigorous statistical inference.

No physical calibration. All measurements are in pixel units. Incorporating pixel-to-micrometer conversion from microscope metadata would yield physically meaningful areas and speeds.

Limited ground-truth annotations. Pixel-level validation was performed on only 5 images. A larger annotated dataset would provide tighter estimates of segmentation accuracy.

Slow myoblast closure. The modest closure observed ($<20\%$ in 24 h) limits the statistical power of dose-response comparisons. The narrow dynamic range may mask pharmacological effects that would become apparent over longer observation windows.

4.5 Future Directions

Planned improvements include: (1) adaptive per-column thresholding using local statistics or Otsu’s method; (2) sub-pixel edge extraction via parabolic fitting of the variance profile around the threshold crossing; (3) bootstrap-based uncertainty quantification for area measurements; (4) integration of pixel-size metadata for calibrated physical measurements; and (5) extension to non-horizontal wound geometries via principal-axis rotation.

From a biological perspective, mathematical models such as the Fisher–Kolmogorov reaction-diffusion framework [61–63] could be applied to the Quantification-derived wound area trajectories to disentangle the contributions of cell migration and proliferation to wound closure, providing mechanistic insight beyond the aggregate closure rate.

5 Conclusion

We have presented Quantification, a fully automated pipeline for scratch-wound healing quantification that combines variance-based single-frame detection with a biologically motivated monotonic closure constraint. The pipeline requires no training data, processes standard phase-contrast microscopy images in linear time, and produces smooth, strictly non-increasing wound area trajectories. Validation on 325 images from a pharmacological study in immortalized myoblasts demonstrated 98.9% detection accuracy and a mean Dice coefficient of 0.91 against expert annotations. Application to 282 validated trajectories revealed that ALK5 inhibition suppressed wound closure kinetics while candesartan had no appreciable effect, consistent with a dominant role for canonical TGF- β signaling in myoblast migration. A three-tier quality-control system ensures robust performance, and parallel trajectory processing enables scalable analysis of high-content experiments. The mathematical formulation is presented in full, all parameters are documented, and the implementation is released as open-source Python code. We anticipate that Quantification will reduce the time and subjectivity currently associated with scratch-wound quantification, enabling researchers to focus on biological interpretation rather than manual measurement.

Data and Code Availability. The Quantification segmentation pipeline is implemented in Python 3.11 and is available at [repository URL]. Raw microscopy images and processed quantitative data are available at [data repository]. All analysis scripts, configuration files, and parameter documentation are included in the repository.

References

- [1] Chung-Chi Liang, Andrea Y Park, and Jun-Lin Guan. In vitro scratch assay: a convenient and inexpensive method for analysis of cell migration in vitro. *Nature Protocols*, 2(2):329–333, 2007. doi: 10.1038/nprot.2007.30.
- [2] James E A Jonkman, Jenny A Cathcart, Fuchun Xu, et al. An introduction to the wound healing assay using live-cell microscopy. *Cell Adhesion and Migration*, 8(5):440–451, 2014. doi: 10.4161/cam.36224.
- [3] Adnan Grada, Marta Otero-Vinas, Francisco Prieto-Castrillo, et al. Research techniques made simple: analysis of collective cell migration using the wound healing assay. *Journal of Investigative Dermatology*, 137(2):e11–e16, 2017. doi: 10.1016/j.jid.2016.11.020.
- [4] Julie C Yarrow, Zachary E Perlman, Nicholas J Westwood, and Timothy J Mitchison. A high-throughput cell migration assay using scratch wound healing, a comparison of image-based readout methods. *BMC Biotechnology*, 4:21, 2004. doi: 10.1186/1472-6750-4-21.
- [5] William J Ashby and Andries Zijlstra. Established and novel methods of interrogating two-dimensional cell migration. *Integrative Biology*, 4(11):1338–1350, 2012. doi: 10.1039/c2ib20154b.
- [6] Beatriz I Pinto et al. Comparison of in vitro scratch wound assay experimental procedures. *Biochemistry and Biophysics Reports*, 33:101423, 2023. doi: 10.1016/j.bbrep.2023.101423.
- [7] Agathe Stamm, Kerstin Reimers, et al. In vitro wound healing assays — state of the art. *BioNanoMaterials*, 17(1-2):79–87, 2016. doi: 10.1515/bnm-2016-0002.
- [8] Tobias Gebäck, Marcel M P Schulz, Petros Koumoutsakos, and Michael Detmar. TScratch: a novel and simple software tool for automated analysis of monolayer wound healing assays. *BioTechniques*, 46(4):265–274, 2009. doi: 10.2144/000113083.
- [9] Alejandra Suarez-Arnedo, Felipe Torres Figueroa, Carolina Clavijo, et al. An ImageJ plugin for the high throughput image analysis of in vitro scratch wound healing assays. *PLOS ONE*, 15(7):e0232565, 2020. doi: 10.1371/journal.pone.0232565.
- [10] Alexander Vargas, Michael Angeli, Chiara Pastrello, et al. Robust quantitative scratch assay. *Bioinformatics*, 32(9):1439–1440, 2016. doi: 10.1093/bioinformatics/btv746.
- [11] Marcos Nunes Viana, Luis Felipe Leite Oliveira, Romulo A Nunes, and Eduardo L Santos de Oliveira. AIM: a computational tool for the automatic quantification of scratch wound healing assays. *Applied Sciences*, 7(12):1237, 2017. doi: 10.3390/app7121237.

- [12] Claire McQuin, Allen Goodman, Vasiliy Chernyshev, et al. CellProfiler 3.0: next-generation image processing for biology. *PLOS Biology*, 16(7):e2005970, 2018. doi: 10.1371/journal.pbio.2005970.
- [13] Simone Martinotti et al. A novel method for evaluating and visualizing scratch wound healing assays using level-set and image sector analysis. *PNAS Nexus*, 4(11):pgaf355, 2025. doi: 10.1093/pnasnexus/pgaf355.
- [14] Katrina K Treloar and Matthew J Simpson. Sensitivity of edge detection methods for quantifying cell migration assays. *PLOS ONE*, 8(6):e67389, 2013. doi: 10.1371/journal.pone.0067389.
- [15] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.
- [16] Devran Doğru, Gökhan D Özdemir, Mehmet A Özdemir, and Utku K Ercan. An automated in vitro wound healing microscopy image analysis approach utilizing U-net-based deep learning methodology. *BMC Medical Imaging*, 24:152, 2024. doi: 10.1186/s12880-024-01332-2.
- [17] Majd A Naser et al. Automated segmentation of hepatic tumor cell migration using U-Net models. *Lecture Notes in Computer Science (Springer)*, 2025. doi: 10.1007/978-3-032-09044-7_30.
- [18] Tim Lehmann et al. Virtually objective quantification of in vitro wound healing scratch assays with the Segment Anything Model, 2024.
- [19] Johannes Oldenburg et al. Methodology for comprehensive cell-level analysis of wound healing experiments using deep learning in MATLAB. *BMC Molecular and Cell Biology*, 22:1, 2021. doi: 10.1186/s12860-021-00369-3.
- [20] Anna Haeger, Katarina Wolf, Mirjam M Zegers, and Peter Friedl. Collective cell migration: implications for wound healing and cancer invasion. *Burns and Trauma*, 1(1):21–28, 2013. doi: 10.4103/2321-3868.113331.
- [21] A Ele, K Richter, K Gros, et al. Cellular dynamics of myogenic cell migration: molecular mechanisms and implications for skeletal muscle cell therapies. *EMBO Molecular Medicine*, 12(12):e12357, 2020. doi: 10.15252/emmm.202012357.
- [22] Keith P Goetsch, C Snyman, Kathryn H Myburgh, and Carola U Niesler. In vitro myoblast motility models: investigating migration dynamics for the study of skeletal muscle repair. *Journal of Muscle Research and Cell Motility*, 34(5-6):333–342, 2013. doi: 10.1007/s10974-013-9364-7.
- [23] Marie T Webster and Chen-Ming Fan. c-MET regulates myoblast motility and myocyte fusion during adult skeletal muscle regeneration. *Development*, 140(22):4586–4596, 2013. doi: 10.1242/dev.098210.
- [24] Kamil Kowalski, Magdalena Kolbowicz, Pawel Krzeminski, et al. Stem cells migration during

- skeletal muscle regeneration — the role of Sdf-1/Cxcr4 and Sdf-1/Cxcr7 axis. *Cell Adhesion and Migration*, 11(4):384–398, 2017. doi: 10.1080/19336918.2016.1227911.
- [25] Keith P Goetsch, Kathryn H Myburgh, and Carola U Niesler. Optimization of the scratch assay for in vitro skeletal muscle wound healing analysis. *Analytical Biochemistry*, 411(1):158–160, 2011. doi: 10.1016/j.ab.2010.12.012.
 - [26] Tadashi Yoshida, Alexander M Tabony, Sheridan Galvez, et al. Angiotensin II inhibits satellite cell proliferation and prevents skeletal muscle regeneration. *Journal of Biological Chemistry*, 288(33):23823–23832, 2013. doi: 10.1074/jbc.M112.449074.
 - [27] Tadashi Yoshida, Laura Semprun-Prieto, Sergiy Sukhanov, and Patrice Delafontaine. Regulation of muscle satellite cell activation and chemotaxis by angiotensin II. *PLOS ONE*, 5(12):e15212, 2010. doi: 10.1371/journal.pone.0015212.
 - [28] Diogo R Lemos, Farshad Babaeijandaghi, Mark Low, et al. Angiotensin-II drives human satellite cells toward hypertrophy and myofibroblast trans-differentiation by two independent pathways. *International Journal of Molecular Sciences*, 20(19):4912, 2019. doi: 10.3390/ijms20194912.
 - [29] Ronald D Cohn, Christel van Erp, Jennifer P Habashi, et al. Angiotensin II type 1 receptor blockade attenuates TGF-beta-induced failure of muscle regeneration in multiple myopathic states. *Nature Medicine*, 13(2):204–210, 2007. doi: 10.1038/nm1536.
 - [30] Hamid S Bedair, Thirumaaran Karthikeyan, Alejandro Quintero, et al. Angiotensin II receptor blockade administered after injury improves muscle regeneration and decreases fibrosis in normal skeletal muscle. *American Journal of Sports Medicine*, 36(8):1548–1554, 2008. doi: 10.1177/0363546508315470.
 - [31] Masatsugu Kurosaka, Taku Suzuki, Koji Hosono, et al. Reduced angiogenesis and delay in wound healing in angiotensin II type 1a receptor-deficient mice. *Biomedicine and Pharmacotherapy*, 63(9):627–634, 2009. doi: 10.1016/j.biopha.2009.01.001.
 - [32] Claudio Cabello-Verrugio, Maria G Morales, Juan C Rivera, et al. Renin–angiotensin system: an old player with novel functions in skeletal muscle. *Medicinal Research Reviews*, 35(3):437–463, 2015. doi: 10.1002/med.21343.
 - [33] Ying Li, William Foster, Bridget M Deasy, et al. Transforming growth factor-beta1 induces the differentiation of myogenic cells into fibrotic cells in injured skeletal muscle. *American Journal of Pathology*, 164(3):1007–1019, 2004. doi: 10.1016/s0002-9440(10)63188-4.
 - [34] Ann E Loiselle, Shawn L K Bowers, Brandon T Paul, et al. Transforming growth factor-beta1 induces transdifferentiation of myoblasts into myofibroblasts via up-regulation of sphingosine kinase-1/S1P3 axis. *Molecular Biology of the Cell*, 20(15):3505–3513, 2009. doi: 10.1091/mbc.e09-09-0812.
 - [35] D C McFarland et al. TGF-beta inhibits muscle differentiation through functional repression of

- myogenic transcription factors by Smad3. *Genes and Development*, 15(22):2950–2966, 2001. doi: 10.1101/gad.925901.
- [36] Fabio Girardi et al. TGFbeta signaling curbs cell fusion and muscle regeneration. *Nature Communications*, 12:750, 2021. doi: 10.1038/s41467-020-20289-8.
 - [37] Erkan Ceko and Elizabeth M McNally. Modifying muscular dystrophy through transforming growth factor-beta. *FEBS Journal*, 280(17):4198–4209, 2013. doi: 10.1111/febs.12266.
 - [38] Estelle Lach-Trifilieff et al. Inhibition of activin-like kinase 4/5 attenuates cancer cachexia-associated muscle wasting. *Scientific Reports*, 9:9826, 2019. doi: 10.1038/s41598-019-46178-9.
 - [39] Tobias Weiss, Eva Hildebrandt, Jacopo Gherardini, et al. Effect of transforming growth factor beta receptor I inhibitors on myotube formation in vitro. *Scientific Reports*, 15:87934, 2025. doi: 10.1038/s41598-025-09381-5.
 - [40] Tobias Weiss et al. The effects of TGF-beta receptor I inhibitors on myofibroblast differentiation and myotube formation. *Frontiers in Cell and Developmental Biology*, 13:1636884, 2025. doi: 10.3389/fcell.2025.1636884.
 - [41] Manasi Vishwakarma et al. Leader cell positioning drives wound-directed collective migration in TGFbeta-stimulated epithelial sheets. *Molecular Biology of the Cell*, 25(9):1586–1593, 2014. doi: 10.1091/mbc.e14-01-0697.
 - [42] Juan Rodriguez-Vita et al. Essential role of Smad3 in angiotensin II-induced vascular fibrosis. *Circulation Research*, 96(12):1292–1299, 2005. doi: 10.1161/01.RES.0000171524.42914.37.
 - [43] Weiping Wang, Xiao-Ru Huang, Erwin Canlas, et al. Angiotensin II activates the Smad pathway during epithelial mesenchymal transdifferentiation. *Kidney International*, 74(5):585–595, 2008. doi: 10.1038/ki.2008.302.
 - [44] Aoife M Murphy, Andrew L Wong, and Meera Bhagra. Modulation of angiotensin II signaling in the prevention of fibrosis. *Fibrogenesis and Tissue Repair*, 8:7, 2015. doi: 10.1186/s13069-015-0023-z.
 - [45] Zhaozheng Yin, Takeo Kanade, and Mei Chen. Understanding the phase contrast optics to restore artifact-free microscopy images for segmentation. *Medical Image Analysis*, 16(5):1047–1062, 2012. doi: 10.1016/j.media.2011.12.006.
 - [46] Paulo E Ambrossio, Pablo Enfedaque, and Jurandir C de Oliveira. Automated cell foreground-background segmentation with phase-contrast microscopy images. *Bioengineering*, 9(2):81, 2022. doi: 10.3390/bioengineering9020081.
 - [47] Tomas Vicar et al. Cell segmentation methods for label-free contrast microscopy: review and comprehensive comparison. *BMC Bioinformatics*, 20:121, 2019. doi: 10.1186/s12859-019-2880-8.
 - [48] Jordao Bragantini et al. Ultrack: pushing the limits of cell tracking across biological scales. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02778-0.

- [49] Kwanghun Lee et al. DeepSea is an efficient deep-learning model for single-cell segmentation and tracking in time-lapse microscopy. *Cell Reports Methods*, 3(7):100523, 2023. doi: 10.1016/j.crmeth.2023.100523.
- [50] Kang Li et al. Image segmentation and dynamic lineage analysis in single-cell fluorescence microscopy. *Cytometry Part A*, 73A(5):376–386, 2008. doi: 10.1002/cyto.a.20527.
- [51] Dayan Guan et al. Domain adaptive video segmentation via temporal consistency regularization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. doi: 10.1109/ICCV48922.2021.00766.
- [52] L Koenig et al. Segmentation of phase contrast microscopy images based on multi-scale local Basic Image Features histograms. *Computer Methods in Biomechanics and Biomedical Engineering: Imaging and Visualization*, 5(5):358–368, 2017. doi: 10.1080/21681163.2015.1016243.
- [53] Christos P Loizou et al. Adaptable texture-based segmentation by variance and intensity for automatic detection of semitranslucent and pink blush areas in basal cell carcinoma. *Skin Research and Technology*, 22(4):412–423, 2016. doi: 10.1111/srt.12281.
- [54] Martin A Fischler and Robert C Bolles. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. doi: 10.1145/358669.358692.
- [55] Abraham Savitzky and Marcel J E Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964. doi: 10.1021/ac60214a047.
- [56] Stefan van der Walt et al. scikit-image: image processing in Python. *PeerJ*, 2:e453, 2014. doi: 10.7717/peerj.453.
- [57] Michael D Zordan, Charles P Mill, Daniel J Riese, and James F Leary. A high throughput, interactive imaging, bright-field wound healing assay. *Cytometry Part A*, 79(3):227–232, 2011. doi: 10.1002/cyto.a.21029.
- [58] Kerstin Schmidt, Dennis Lerm, Anna Schmidt, et al. Automated high-throughput live cell monitoring of scratch wound closure. *SLAS Technology*, 2024. doi: 10.1177/11795972241295619.
- [59] Kinga Delaney, Paulina Kasprzycka, Maria A Ciemerych, and Marta Zimowska. The role of TGF-beta1 during skeletal muscle regeneration. *Cell Biology International*, 41(7):706–715, 2017. doi: 10.1002/cbin.10725.
- [60] Gil Topman, Orna Sharabani-Yosef, and Amit Gefen. The frequent sampling of wound scratch assay reveals the “opportunity” window for quantitative evaluation of cell motility-impeding drugs. *Frontiers in Cell and Developmental Biology*, 9:640972, 2021. doi: 10.3389/fcell.2021.640972.
- [61] Wang Jin, Esha T Shah, Catherine J Penington, et al. Estimating cell diffusivity and cell

- proliferation rate by interpreting IncuCyte ZOOM assay data using the Fisher–Kolmogorov model. *BMC Systems Biology*, 9:38, 2015. doi: 10.1186/s12918-015-0182-y.
- [62] Philip K Maini, Donald L S McElwain, and David I Leavesley. Scratch assay microscopy: a reaction-diffusion equation approach for common instruments and data. *Mathematical Biosciences*, 330:108482, 2020. doi: 10.1016/j.mbs.2020.108482.
- [63] Matthew J Simpson, Katrina K Treloar, Benjamin J Binder, et al. Quantifying the roles of cell motility and cell proliferation in a circular barrier assay. *Journal of the Royal Society Interface*, 10(82):20130007, 2013. doi: 10.1098/rsif.2013.0007.

Supplementary Material

Supplementary Tables

Table S1: Sample sizes (number of trajectories) per cell line and treatment condition after QC filtering.

Cell line	Alk5i	DMSO	Media
Line 1	10	8	15
Line 2	5	15	6

Table S2: Sample accounting: initial, QC-passing, and final trajectory counts per group, with exclusion percentages. High exclusion rates in DMSO and Media reflect the 0.1 mM concentration filter (only trajectories at the analysis concentration are retained).

Cell line	Condition	n initial	n QC pass	n final	n excluded	Excl. (%)
Line 1	Alk5i	11	10	10	1	9.1
Line 1	DMSO	24	8	8	16	66.7
Line 1	Media	31	15	15	16	51.6
Line 2	Alk5i	8	5	5	3	37.5
Line 2	DMSO	32	15	15	17	53.1
Line 2	Media	22	6	6	16	72.7

Table S3: Per-frame processing sequence of the temporal integrator ($t > 0$).

Step	Operation	Implemented in
1	Single-frame detector $\rightarrow$ raw edges $u_{\text{det}}(x, t)$, $\ell_{\text{det}}(x, t)$	Section 2.3.2
2	NaN guard: replace non-finite detector outputs with previous state	<code>KalmanEdgeFilter.update</code>
3	Spatial Gaussian smoothing of detected edges ($\sigma_{\text{spatial}} = 5$ columns)	<code>gaussian_filter1d</code>
4	Per-column observation noise $R(x, t)$ estimation	<code>estimate_R_per_column</code>
5	Kalman predict step: $P^-(x, t) = P(x, t - 1) + Q$	<code>_kalman_step</code>
6	Kalman update: gain K , blended edge u_{filt} , contracted P_{upd}	<code>_kalman_step</code>
7	NaN guard on Kalman output: reset non-finite u_{filt} to $u(x, t - 1)$	<code>KalmanEdgeFilter.update</code>
8	Hard safety net: $u(x, t) = \max(u_{\text{filt}}, u(x, t - 1))$	<code>KalmanEdgeFilter.update</code>
9	P-state correction: reset P to P^- for columns where safety net fired	<code>KalmanEdgeFilter.update</code>
10	Mask reconstruction from constrained edges	<code>_edges_to_mask</code>
11	Last-resort area backstop: emit warning and revert if $A(t) > A(t - 1)$	<code>QuantificationSegmenter.segment</code>

Table S4: Descriptive statistics for the primary endpoint (wound closure distance, pixels) and secondary endpoint (closure speed, pixels/h) at the final timepoint, stratified by cell line and treatment condition. Values are mean $\pm$ SEM with 95% confidence intervals.

Cell line	Condition	Distance mean	Distance SD	Distance SEM	CI ₉₅ lower	CI ₉₅ upper	n
Line 1	Alk5i	16.34	8.73	2.76	10.93	21.75	10
Line 1	DMSO	19.28	4.94	1.75	15.86	22.70	8
Line 1	Media	20.87	9.06	2.34	16.28	25.45	15
Line 2	Alk5i	16.56	6.93	3.10	10.49	22.64	5
Line 2	DMSO	21.69	19.40	5.01	11.87	31.51	15
Line 2	Media	18.86	4.02	1.64	15.65	22.08	6

Table S5: QC pass rate and monotonic constraint rate per group. All QC-passing trajectories were constrained, consistent with the biological axiom.

Cell line	Condition	QC pass rate (%)	Constraint rate (%)	n total
Line 1	Alk5i	90.9	100	11
Line 1	DMSO	100.0	100	8
Line 1	Media	100.0	100	15
Line 2	Alk5i	62.5	100	8
Line 2	DMSO	93.8	100	16
Line 2	Media	85.7	100	7

Table S6: Pairwise comparisons of closure speed and distance versus DMSO, stratified by cell line. Welch’s t -test; p -values shown before and after Bonferroni correction (4 tests per metric). All corrected p -values equal 1.000.

Cell line	Condition	Metric	t -stat	Cohen’s d	p (raw)	p (Bonf.)	n_{trt}	n_{ctrl}
Line 1	Media	speed	0.543	0.200	0.593	1.000	15	8
Line 1	Alk5i	speed	−0.900	−0.402	0.382	1.000	10	8
Line 2	Media	speed	−0.537	−0.169	0.599	1.000	6	15
Line 2	Alk5i	speed	−0.871	−0.294	0.396	1.000	5	15
Line 1	Media	distance	0.543	0.200	0.593	1.000	15	8
Line 1	Alk5i	distance	−0.900	−0.402	0.382	1.000	10	8
Line 2	Media	distance	−0.537	−0.169	0.599	1.000	6	15
Line 2	Alk5i	distance	−0.871	−0.294	0.396	1.000	5	15

Table S7: Mixed-effects model parameter estimates. Model: distance $\sim$ time $\times$ condition $\times$ cell_line, with random intercept and random slope per trajectory. The main effect of time is highly significant ($p < 10^{-12}$); no interaction terms reach significance.

Term	Estimate	p -value	CI ₉₅ lower	CI ₉₅ upper
Intercept	0.847	0.744	−4.237	5.931
DMSO vs. Alk5i	0.864	0.828	−6.953	8.681
Media vs. Alk5i	1.378	0.686	−5.305	8.060
Line 2 vs. Line 1	1.710	0.670	−6.164	9.585
DMSO $\times$ Line 2	2.851	0.602	−7.873	13.574
Media $\times$ Line 2	−2.239	0.691	−13.268	8.789
t (main effect)	0.879	<0.001	0.639	1.120
$t \times$ DMSO	0.143	0.438	−0.218	0.504
$t \times$ Media	0.161	0.309	−0.149	0.472
$t \times$ Line 2	−0.122	0.563	−0.536	0.292
$t \times$ DMSO $\times$ Line 2	−0.065	0.810	−0.597	0.466
$t \times$ Media $\times$ Line 2	−0.019	0.945	−0.573	0.534
<i>Variance components</i>				
Group variance	8.278	<0.001	5.144	11.411
Group $\times t$ covariance	0.096	0.063	−0.005	0.197
t variance	0.016	<0.001	0.009	0.023

Table S8: Statistical assumption checks. Shapiro–Wilk normality test for wound closure distance, per group. Levene’s test for homogeneity of variance across conditions, per cell line. DMSO Line 2 shows significant deviation from normality ($p < 0.001$), warranting use of Welch’s t -test (unequal variances assumed).

Cell line	Condition	Shapiro W	Shapiro p	n
Line 1	Alk5i	0.961	0.799	10
Line 1	DMSO	0.915	0.393	8
Line 1	Media	0.957	0.634	15
Line 2	Alk5i	0.780	0.055	5
Line 2	DMSO	0.551	<0.001	15
Line 2	Media	0.876	0.253	6
<i>Levene’s test (homogeneity of variance across conditions)</i>				
Line 1		$F = 1.457, p = 0.249$		
Line 2		$F = 0.412, p = 0.667$		

Table S9: Inter-experiment reproducibility: mean wound closure distance (pixels) at the final timepoint, by cell line, condition, and experiment. Experiment 2 data were unavailable for some condition–cell line combinations (see text for details).

Cell line	Condition	Mean EXP1	Mean EXP2	Difference	n EXP1	n EXP2
Line 1	Alk5i	10.94	17.69	-6.75	2	8
Line 1	DMSO	19.28	—	—	8	0
Line 1	Media	17.63	23.70	-6.07	7	8
Line 2	Alk5i	16.56	—	—	5	0
Line 2	DMSO	25.82	16.98	8.84	8	7
Line 2	Media	18.86	—	—	6	0